

POWER PLANT DESIGN

LAB 2

OBJECTIVES

- ❖ To select the number of units and the installed capacity of each unit (MW) with reasons.
- ❖ To calculate the monthly energy generation and annual energy generation in MWh.
- ❖ To calculate the maximum annual energy generation by assuming the design discharge is available throughout the year and calculate annual plant factor.
- ❖ To select the appropriate turbine for each unit with supporting calculations and reference documents.

PROCEDURE

From the lab 1 the designed values are:

- ❖ Hydropower Capacity (P_{design}) = 273.42 MW
- ❖ Net head (H_{net}) = 530.1m

This head was calculated after setting the intake at 27°55'35.72"N 85°56'4.90"E and the power house at 27°46'0.58"N 85°52'44.19"E. The intake was chosen at that location based on NITI foundation data. The powerhouse was chosen so to minimize the distance from prebuilt transmission network to connect to national grid. The head calculated from Google Earth Pro by estimating a realistic path as shown in Lab 1. In real practice, survey gives water elevation level between intake and tailrace from which we calculate actual head which is usually lower than the value from rough estimation.

- ❖ Design Discharge ($Q_{45\%}$) = 52.578m³/s

For discharge calculation, NEA 1997 method was used. The mean ppt was calculated based on Roll No. as instructed in Lab 1. The catchment area was calculated based on the QGIS software which is standard practice in the industry as shown in Lab 1. In actual practice for large hydropower project >100 MW such as this, a stream gauging site has to be set to give long term flow data using velocity area method to find real variation in discharge.

1 Plant factor calculation

The Electrical Power Output and Monthly Energy Generation all the months are calculated and their values are tabulated as below:

Month	mean monthly flow(m ³ /s)	Actual Discharge(m ³ /s) (Mean monthly flow - riparian) limited to Q _{design}	Power (MW)	Monthly Energy Generation (GWh)
January	23.32859815	21.45650937	97.40923348	72.47246971
February	19.64892356	17.77683478	80.70408006	55.20159076
March	18.72088783	16.84879905	76.49094139	56.90926039
April	26.86646285	24.99437407	113.470592	81.69882621
May	19.39176516	17.51967637	79.53662067	59.17524578
June	108.176463	52.57806758	238.6962937	171.8613314
July	318.4661801	52.57806758	238.6962937	177.5900425
August	363.2912246	52.57806758	238.6962937	177.5900425
September	273.9549109	52.57806758	238.6962937	171.8613314
October	120.2405487	52.57806758	238.6962937	177.5900425
November	57.98939313	52.57806758	238.6962937	171.8613314
December	38.0214277	36.14933892	164.1124069	122.0996307
			Total GWh	1495.911145
			After 5% outage	1421.115588

The maximum annual energy that can be generated is given by,

$$E_{\max} = .90 \times .97 \times 52.578 \times 9.81 \times 530.1 \times 24 \times 30 \times 12 / 10^6 \\ = 2062.35 \text{ Gwh}$$

Now, Annual **Plant Factor calculation**

$$\text{Annual plant factor} = E_{\text{act}} / E_{\max} \\ = 1421.12 / 2062.33 = 0.689$$

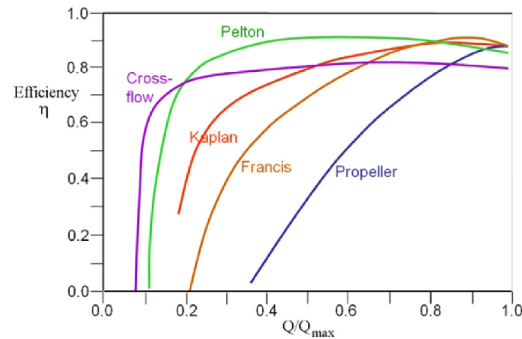
2. Selection of number of units and the installed capacity of each unit (MW, MVA) with reasons.

Multiple no. of units is considered for the analysis. For each alternative, part flow efficiency was calculated in the driest condition. As the no. of units increases the part flow in the turbine for the worst case also increases increasing efficiency in that part flow condition. But increasing no of units also increases initial cost of the project. But, the annuity (annual income) also increases due to the increase in efficiency in dry conditions. So, financial analysis is done for all the technically viable alternatives. By comparing the discounted PB period, ERR, IRR and NPV of the project the best alternative is selected. Multiple units are usually preferred due to increase in reliability i.e., downtime decreasing due to failure of units, not having to completely shut down plant for maintenance purposes.

For this lab however financial analysis is not done as it is a lengthy process, so only part flow efficiency is considered in the worst case (driest month of March) and compared. Then the RPM is calculated for that case to see if there is a generator with suitable no. of poles.

In March (driest month), the discharge is 16.8 cumec.

The part flow efficiency of various turbines as shown in figure below. Due to high head of the project only Pelton wheel is considered.



No. of units	Design Discharge per turbine(Q45)	No. of units active in march(n)	Actual Q in march(Q)	Part flow in March(Q/Q45)	Part flow efficiency	Nsyn	No of poles
3	17.52	1	16.8	0.96	0.8	233	26
4	13.14	1	13.14	1	0.87	269	22
5	10.51	2	8.4	0.8	0.85	301	20
6	8.76	2	8.4	0.91	0.83	330	18

As seen in part flow efficiency curve and the table, Pelton wheel turbine has great performance in a big range of part flow condition. So, 5 no of units were selected as it gives great part flow efficiency for all months and there is no wastage of available discharge in all months.

Let us select five units each of capacity 47.739 MW. In six months where sufficient flow is available all five unit can be operated but in dry season either two or three can be operated while other can be sent for maintenance.

For example, in March (driest month), the discharge is 16.8 cumec. So, if we select 5 units the discharge per unit is 10.51 cumec. So, part flow efficiency is maintained to acceptable level. Five units were also selected to find generator with suitable no. of poles that is available.

No. of units = 5

Total installed capacity = 238.696 MW

In context of Nepal, power factor is taken as 0.85, so

MVA rating of each unit = $47.739/0.85 = 56.16$ MVA

Total MVA = $5 \times 56.16 = 280.819$ MVA

Design discharge per unit = $52.57/5 = 10.515 \text{ m}^3/\text{s}$.

3. Turbine Selection

The head Ranges for some of the turbines are;

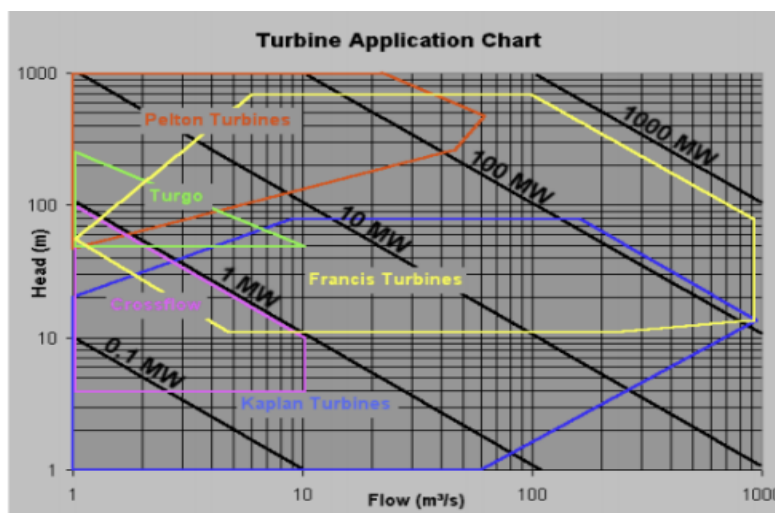
(a) Kaplan turbine: $2 \leq H \leq 70$

(b) Francis Trubine: $10 \leq H \leq 300$

(c) Pelton Turbine : $80 \leq H \leq 1600$

Here, Kaplan and Francis turbine is discarded due to insufficient head. The choice is Pelton turbine.

Alstom chart:



From the Alstom chart, it is observed that, for net head of 530m, PU discharge of $10.514 \text{ m}^3/\text{s}$ the **Pelton turbine is in the range.**

Therefore, I have chosen Pelton Turbine for all five generating units.

Specific speed (N_s)

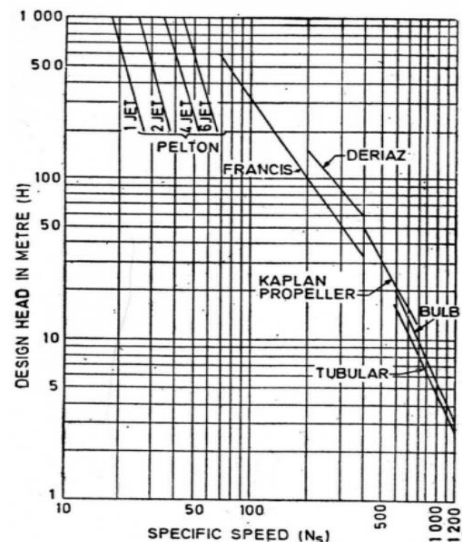
Specific speed is specified as the speed in revolutions per minute at which the given turbine would rotate, if reduced homologally in size, so that it would develop one metric horse power at full gate opening under one meter head.

$$N_s = \frac{NP^{0.5}}{H^{1.25}}$$

Where, N = Synchronous speed in RPM

P = Power in HP

H = Net head



From this graph it is observed that, for a net head of 530m for Pelton turbine (2 jet), $N_s = 30$ RPM

Using the above relation to obtain synchronous speed N_{syn}

$$N_{syn} = 301.58 \text{ RPM}$$

Number of Poles	RPM (50 Hz)	Number of Poles	RPM (50Hz)	Number of Poles	RPM (50Hz)
2	3000	2	3000	16	375
4	1500	4	1500	18	333
6	1000	6	1000	20	300
8	750	8	750	22	272
10	600	10	600	24	250
12	500	12	500	26	230
14	428	14	428	28	214

Therefore, the nearest standard synchronous speed 300 RPM with 20 generator poles is selected.

Turbine Dimension:

For a **2-jet Pelton Turbine** given head H , flow Q , synchronous speed N , specific speed N_s :

1. Jet velocity: $V_j = C_v \sqrt{2gH} = 98.92 \text{ m/s}$, where $C_v = 0.97$
2. Runner diameter: $D = \frac{60 K_u \sqrt{2gH}}{\pi N} = 2.9 \text{ m}$, where $K_u = 0.45$ for maximum efficiency
3. Jet diameter: $d_j = \sqrt{\frac{4Q}{2\pi V_j}} = 0.29 \text{ m}$
4. Jet ratio: $m = D/d_j = 11$
5. Number of buckets: $Z \approx 15 + D/2d_j = 21$ buckets
6. Bucket breadth: $B \approx 4d_j = 1.04 \text{ m}$
7. Bucket length: $L \approx 2.5 d_j = 0.65 \text{ m}$
8. Bucket radial depth: $\approx d_j = 0.26 \text{ m}$

RESULTS/DISCUSSION

The hydrological assessment based on Q45% design discharge guided the division of installed capacity into multiple generating units to enhance operational flexibility and minimize dry-season spillage.

Monthly energy analysis reveals a strong seasonal trend, with peak generation during the monsoon and reduced output in the dry season. The annual energy generation, obtained from monthly summation, was used to evaluate the plant factor against the maximum possible energy assuming continuous design discharge. For a net head of 530 m, the calculated specific speed confirms the suitability of a Pelton turbine. Overall, the selected configuration ensures high efficiency, hydraulic safety, and reliable performance for the Run-of-River scheme.

CONCLUSION

The five-unit configuration enhances energy recovery, reduces dry-season spillage, and improves annual generation, thereby justifying the design choice. Overall, the results demonstrate that the proposed hydropower scheme is feasible, efficient, and well aligned with the site's hydraulic and operational conditions.